

DATA DICTIONARY

& RELATIONSHIP REFERENCE

Healthcare Claims & Hospital Operations Analytics (HCHOP)

Version 1.0 · June 2024 · Data Engineering Team

1

CMS Medicare Part D — Drug
Prescribers by Provider & Drug

CSV 50,000 rows

One row = one prescriber (identified by NPI) for one drug. All monetary values in USD. Rows where beneficiary count is less than 11 are suppressed by CMS privacy rules — the flag column will show '*'.
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Column Name	Type	Sample Value	Meaning & Notes
Prescriber Identity			
Prscrbr_NPI	PK	1043321819	National Provider Identifier — 10-digit unique ID for every US prescriber. Primary key for this dataset. Use this to link to physician dimension tables.
Prscrbr_Last_Org_Name	VARCHAR	SILVA	Prescriber surname or organisation name (for group practices). Stored as ALL CAPS in source.
Prscrbr_First_Name	VARCHAR	Nicholas	Prescriber given name. Empty for organisations.
Prscrbr_City	VARCHAR	East Kenneth	City where the prescriber's practice is located.
Prscrbr_State_Abrvtn	VARCHAR	AZ	Two-letter US state abbreviation of the prescriber's practice address.
Prscrbr_State_FIPS	VARCHAR	42	FIPS numeric state code. Alternative to state abbreviation for geographic joins.
Prscrbr_Type	VARCHAR	Internal Medicine	Medical specialty derived from claims. E.g. Cardiology, Family Practice, Oncology. Use for specialty-level drug cost analysis.

Prscrbr_Type_Src	VARCHAR	Claim-Specialty	Source used to determine specialty: 'Claim-Specialty' (from claim form) or 'Medicare Specialty Code'. Affects reliability of Prscrbr_Type.
Drug Identity			
Brnd_Name	VARCHAR	Gabapentin	Brand (trade) name of the drug as listed on the claim. When brand name = generic name, no branded product was prescribed.
Gnrc_Name	VARCHAR	Gabapentin	Generic (chemical/INN) name of the drug. Compare with Brnd_Name to detect generic vs branded prescribing. Key for savings analysis.
Total Claims — All Beneficiaries			
Tot_Clms	NUMBER	778	Total number of prescription claims filled by this prescriber for this drug in the year. Includes original fills and refills.
Tot_30day_Fills	NUMBER	803.1	Total 30-day equivalent fills — standardises claims of different days supply. One 90-day fill = 3.0 × 30-day fills.
Tot_Day_Suply	NUMBER	21734	Total days of drug supply dispensed. Divide by Tot_Clms to get average days per prescription.
Tot_Drug_Cst	NUMBER	8729074.38	Total drug cost paid (in USD) by Medicare Part D and beneficiary cost-sharing combined. Primary cost metric for spend analysis.
Tot_Benes	NUMBER	321	Number of unique Medicare beneficiaries who received this drug from this prescriber. Minimum 11 — rows below this are suppressed.
Age 65+ Subset (GE65) — Same metrics for beneficiaries aged ≥ 65			
GE65_Sprsn_Flag	FLAG/CHAR		** if GE65 beneficiary count is < 11 and data is suppressed for privacy. Empty string means data is shown.
GE65_Tot_Clms	NUMBER	380	Claims for beneficiaries aged 65 and over only. Subset of Tot_Clms.
GE65_Tot_30day_Fills	NUMBER	376.9	30-day equivalent fills for beneficiaries aged 65 and over.
GE65_Tot_Drug_Cst	NUMBER	4263558.18	Drug cost for beneficiaries aged 65 and over. Compare with Tot_Drug_Cst to see elderly patient share.
GE65_Tot_Day_Suply	NUMBER	10239	Days of supply for beneficiaries aged 65 and over.

GE65_Bene_Sprsn_Flag	FLAG/CHAR		** if GE65 beneficiary count suppressed. Same logic as GE65_Sprsn_Flag.
GE65_Tot_Benes	NUMBER	253	Number of unique beneficiaries aged 65 and over who received this drug from this prescriber.

2

Synthea Patient Data — HL7 FHIR R4 JSON Bundles

JSON

557 files ·
409,973
entries

Each file is one FHIR R4 Bundle containing ALL resources for a single patient. Resources are stored inside an 'entry' array. Use LATERAL FLATTEN in Snowflake to extract individual resource types. Snowflake colon notation accesses nested fields: BUNDLE_RAW:entry[0]:resource.id::VARCHAR

Resource 1 — Patient

Field Path	Type	Sample Value	Meaning & Notes
resource.id	PK	faac724a-a9e9-b e66-fe1e-3044dc 0ba8ea	UUID uniquely identifying this patient. Natural key for all patient joins. Referenced by subject.reference in other resources.
resource.gender	VARCHAR	male	Patient biological sex: 'male', 'female', 'other', 'unknown'.
resource.birthDate	DATE	1955-02-14	Patient date of birth in YYYY-MM-DD format. Derive age and age_band from this.
resource.address[0].city	VARCHAR	Boston	City of patient's primary address. First element [0] of address array.
resource.address[0].state	VARCHAR	Massachusetts	Full state name (not abbreviation) for US patients.
resource.name[0].family	VARCHAR	Watsica258	Patient surname. Synthea appends random numbers to names — strip numeric suffix for display.
resource.name[0].given[0]	VARCHAR	Brandon214	Patient given name. Same numeric suffix convention.
resource.maritalStatus.coding[0].code	VARCHAR	S	Marital status code: S=Single, M=Married, D=Divorced, W=Widowed.
resource.extension[0](race)	VARCHAR	White	Race from us-core-race extension. Located in extension array — filter by URL 'us-core-race'.

Resource 2 — Encounter

Field Path	Type	Sample Value	Meaning & Notes
resource.id	PK	4a189fae-f542-a234-df39-d9d7b9d2f618	UUID for this encounter. Referenced by encounter.reference in Condition, Procedure, Claim.
resource.status	VARCHAR	finished	'finished' = completed visit. Also: 'in-progress', 'cancelled'. Filter to 'finished' for analytics.
resource.class.code	VARCHAR	AMB	Encounter class: AMB=ambulatory outpatient, IMP=inpatient, EMER=emergency, HH=home health.
resource.type[0].coding[0].display	VARCHAR	Well child visit	Human-readable encounter type description. Use for encounter classification.
resource.subject.reference	FK	urn:uuid:faac7...	Patient UUID with 'urn:uuid:' prefix. Strip this prefix to join to Patient.id.
resource.period.start	TIMESTAMP	2020-12-04T03:26:20-05:00	Encounter start date-time with timezone offset. Cast to TIMESTAMP_NTZ for Snowflake.
resource.period.end	TIMESTAMP	2020-12-04T03:41:20-05:00	Encounter end date-time. Difference from start = length of stay in minutes.
resource.serviceProvider.display	VARCHAR	PCP198892	Name/ID of the hospital or clinic providing care.
resource.participant[0].individual.reference	FK	Practitioner/abc123	Practitioner UUID. Join to Practitioner resource for physician details.

Resource 3 — Condition (Diagnosis)

Field Path	Type	Sample Value	Meaning & Notes
resource.id	PK	2ced594b-2027-47f7-7dec-b225496fa0ed	UUID for this condition record.

resource.subject.reference	FK	urn:uuid:faac7...	Patient UUID. Strip 'urn:uuid:' to join to Patient.id.
resource.encounter.reference	FK	urn:uuid:31d3...	Encounter UUID where this diagnosis was recorded.
resource.code.coding[0].code	VARCHAR	267036007	SNOMED-CT diagnosis code. Note: Synthea uses SNOMED; real claims use ICD-10. Map SNOMED → ICD-10 using a crosswalk table if needed.
resource.code.coding[0].display	VARCHAR	Dyspnea (finding)	Human-readable diagnosis name.
resource.clinicalStatus.coding[0].code	VARCHAR	resolved	Current clinical status: 'active', 'resolved', 'inactive', 'remission'.
resource.onsetDateTime	TIMESTAMP	2021-07-04T04:26:20	When the condition first appeared. Cast to DATE for analytics.
resource.abatementDateTime	TIMESTAMP	2021-08-01T04:26:20	When the condition resolved. NULL if still active.

Resource 4 — Claim

Field Path	Type	Sample Value	Meaning & Notes
resource.id	PK	1a6b1912-3b79-04cf-b1ec-5c67d8546f52	UUID for this claim.
resource.status	VARCHAR	active	'active' = approved/paid. 'cancelled' = denied/voided. Primary filter for FACT_CLAIMS.
resource.patient.reference	FK	urn:uuid:faac7...	Patient UUID. Strip 'urn:uuid:' prefix to join to Patient.id.
resource.created	TIMESTAMP	2020-12-04T03:41:20	Date the claim was submitted. Use as CLAIM_DATE in FACT_CLAIMS.
resource.total.value	NUMBER	240.00	Total billed amount in USD. This is the billed charge, not the paid amount.

resource.provider.display	VARCHAR	PCP198892	Name/ID of the billing provider (hospital or clinic).
resource.facility.reference	FK	Location?id=...	Location UUID of the facility where care was delivered.
resource.diagnosis[0].diagnosisCodeableConcept.coding[0].code	VARCHAR	267036007	Primary diagnosis code on the claim. Links to Condition resource.
resource.insurance[0].coverage.reference	FK	Coverage?id=...	Insurance coverage UUID. Join to Coverage resource for payer details.
resource.item[*].productOrService.coding[0].code	VARCHAR	410620009	Line-item procedure/service code. Array — flatten for per-procedure analysis.
resource.billablePeriod.start	TIMESTAMP	2020-12-04T03:26:20	Start of the service period covered by this claim.
resource.billablePeriod.end	TIMESTAMP	2020-12-04T03:41:20	End of the service period. Difference = length of billed episode.

Resource 5 — MedicationRequest

Field Path	Type	Sample Value	Meaning & Notes
resource.id	PK	med-uuid-001	UUID for this prescription request.
resource.subject.reference	FK	urn:uuid:faac7...	Patient UUID who received the prescription.
resource.encounter.reference	FK	urn:uuid:31d3...	Encounter at which the medication was prescribed.
resource.medicationCodeableConcept.text	VARCHAR	Lisinopril 10 MG	Full drug name including dose. Use for matching to drug spend dataset.
resource.authoredOn	TIMESTAMP	2021-07-04	Date the prescription was written.
resource.dosageInstruction [0].text	VARCHAR	Take 1 tablet daily	Free-text dosage instructions.

resource.status	VARCHAR	active	'active'=ongoing, 'stopped'=discontinued, 'completed'=course finished.
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3

CMS Hospital Provider Cost Report — FY2022

CSV

5,000
hospitals

One row = one hospital for one fiscal year. All revenue and expense amounts are in USD. Fiscal year varies by hospital — not all hospitals have a January–December year. The Provider CCN is the stable natural key across years.

Column Name	Type	Sample Value	Meaning & Notes
Report Identity			
rpt_rec_num	PK	700001	Internal CMS report record number. Unique per row in this file.
Provider CCN	FK	939928	CMS Certification Number — the stable 6-digit identifier for a hospital across all CMS datasets. Natural key for SCD2 joins. Links this file to Medicare claims data.
Facility Location			
Facility Name	VARCHAR	ALLEN-LOPEZ MEDICAL CENTER	Hospital's official CMS-registered name. ALL CAPS in source — apply INITCAP() in Silver.
Street Address	VARCHAR	09793 BOYD PLAINS	Hospital street address. Used for geocoding.
City	VARCHAR	EAST JACK	Hospital city.
State Code	VARCHAR	WA	Two-letter US state abbreviation.
Zip Code	VARCHAR	45002	5-digit ZIP code. Use for geographic clustering.
County	VARCHAR	LAKE STEPHANIEBURY COUNTY	County name. Useful for rural/urban analysis.
Medicare CBSA Number	NUMBER	37493	Core-Based Statistical Area code — CMS's geographic market unit. Hospitals in the same CBSA compete for patients.
Rural versus Urban	VARCHAR	R	'R' = rural, 'U' = urban. Affects Medicare payment rates and benchmarks.
Fiscal Period			
Fiscal Year Begin Date	DATE	02/01/2022	First day of the hospital's fiscal year (MM/DD/YYYY format). Cast with TRY_TO_DATE(col,'MM/DD/YYYY').

Fiscal Year End Date	DATE	01/28/2023	Last day of the hospital's fiscal year. Note: not always 12 months.
Type of Control	VARCHAR	Proprietary - Corporation	Ownership type: 'Voluntary Non-profit - Church', 'Voluntary Non-profit - Other', 'Proprietary - Corporation', 'Government - State', 'Government - Local'. Key for SCD2 tracking.
Capacity & Utilisation			
Number of Beds	NUMBER	139	Licensed beds — the maximum beds the hospital is approved to operate. SCD2 tracked attribute — changes when a hospital expands or reduces capacity.
Total Bed Days Available	NUMBER	50735	Beds × days in fiscal year = maximum possible bed-days. Used to calculate occupancy rate.
Total Days Title XVIII	NUMBER	11624	Inpatient days covered by Medicare (Title XVIII). Divide by Total Days Total for Medicare patient share.
Total Days Total	NUMBER	26877	Total inpatient days across all payers. Divide by Total Bed Days Available = occupancy rate.
Total Discharges Title XVIII	NUMBER	2354	Number of Medicare patient discharges.
Total Discharges Total	NUMBER	5444	Total patient discharges across all payers. Divide Total Days Total by this = average length of stay (ALOS).
Revenue & Financials			
Inpatient Revenue	NUMBER	406711895.32	Gross charges for inpatient services (before contractual adjustments). USD.
Outpatient Revenue	NUMBER	382862143.0	Gross charges for outpatient services. USD.
Total Patient Revenue	NUMBER	789574038.32	Sum of Inpatient + Outpatient Revenue. Total gross charges.
Less Contractual Allowance and Discounts	NUMBER	412982667.54	Amount discounted off gross charges per payer contracts. Subtract from Total Patient Revenue to get Net Patient Revenue.
Net Patient Revenue	NUMBER	376591370.78	Revenue actually collected — Total Patient Revenue minus contractual discounts. Primary top-line revenue metric.
Less Total Operating Expense	NUMBER	282018911.23	Total costs to run the hospital: labour, supplies, drugs, overhead.
Net Income from Service to Patients	NUMBER	94572459.55	Operating profit from patient care: Net Patient Revenue minus Total Operating Expense.

Net Income	NUMBER	94572459.55	Overall net income including non-patient revenue (grants, investments). Often equals Net Income from Service if no other sources.
Medicare Payment Components			
DRG Amounts Other Than Outlier Payments	NUMBER	21622674.75	Base Medicare DRG (Diagnosis Related Group) payments received. Core inpatient Medicare reimbursement.
Total IME Payment	NUMBER	0	Indirect Medical Education payment — bonus paid to teaching hospitals for training residents. Non-zero only for academic medical centres.
Allowable DSH Percentage	NUMBER	0.0606	Disproportionate Share Hospital percentage — fraction of low-income patients served. Higher values = higher safety-net status.
Cost To Charge Ratio	NUMBER	0.4982	Hospital's cost divided by charges. Multiply any charge by this ratio to estimate actual cost. Values < 1 mean charges > costs (typical). Use for cost benchmarking.

4

ICD-10-CM Diagnosis Codes 2025 — Reference File

TXT

141 codes
(sample set)

Tab-delimited text file. No header row. Two fields per line: code then description. Codes follow a strict hierarchical structure. The first character is the chapter letter; subsequent characters drill into sub-categories. This file is a lookup/reference table only — it does not change unless CMS issues a new annual release.

Field	Type	Sample Value	Meaning & Notes
Column 1 — Code	PK	E1100	ICD-10-CM diagnosis code. Length 3–7 characters. No dots in the file (CMS format) — add a dot after position 3 for display: E11.00. Primary key and join key to all claims data.
Column 2 — Description	VARCHAR	Type 2 diabetes mellitus with hyperosmolarity without nonketotic HHNC	Full official CMS description of the diagnosis. Can be up to 200+ characters.

ICD-10 Code Structure Explained

Code pattern	Example	What it means
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First letter	A=Infectious · E=Endocrine I=Circulatory · J=Respiratory K=Digestive · M=Musculoskeletal N=Genitourinary · R=Symptoms S=Injury · Z=Factors	ICD-10 chapter. First letter of the code tells you the disease category. Use LEFT(ICD10_CODE,1) in SQL to group by chapter.
First 3 chars	E11 = Type 2 Diabetes I10 = Hypertension J44 = COPD I50 = Heart Failure	Category (3-char code). The broadest clinically meaningful grouping. Use LEFT(ICD10_CODE,3) in SQL to group by disease category without drilling into complications.
4-7 chars	E1100 = Type 2 diabetes with hyperosmolarity I5032 = Chronic diastolic heart failure	Subcategory — adds specificity: which complication, which body side, which severity. Billable codes are typically 5–7 characters. Use full code for exact diagnosis; use truncated LEFT(code,3) for grouping.

Relationships — Join Keys & Table Links

This section defines every relationship between the source files (Bronze) and the Silver/Gold tables. Each row shows the key column on both sides of the join, the cardinality, and what the join enables.

Source-to-Source Relationships (cross-file joins)

From Table	Key Column	Cardinality	To Table	Key Column	Enables
RAW_DRUG_SPEND	Prscrbr_NPI	M : 1	DIM_PHYSICIAN (Silver)	npi	Prescriber details
RAW_DRUG_SPEND	Brnd_Name + Gnrc_Name	M : 1	DIM_DRUG (Silver)	brand + generic	Drug dimension
RAW_PATIENT_FHIR	patient.id	1 : M	RAW_PATIENT_FHIR (Claim)	patient.reference	All claims for patient
RAW_PATIENT_FHIR	encounter.id	1 : M	RAW_PATIENT_FHIR (Condition)	encounter.reference	Diagnoses per visit
RAW_PATIENT_FHIR	claim.diagnosis.code	M : 1	REFERENCE.ICD10_CODES	ICD10_CODE	Readable diagnosis name
RAW_HOSPITAL_COST	Provider CCN	1 : M	RAW_DRUG_SPEND (via NPI→CCN)	Prscrbr facility	Cost per prescriber

Silver Dimension → Fact Relationships (Star Schema)

Dimension Table	Key Column	Cardinality	Fact Table	FK Column	Notes
DIM_PATIENT	PATIENT_SK	1 : M	FACT_CLAIMS	PATIENT_SK	Always join IS_CURRENT=TRUE
DIM_FACILITY	FACILITY_SK	1 : M	FACT_CLAIMS	FACILITY_SK	Always join IS_CURRENT=TRUE
DIM_DIAGNOSIS	DIAG_SK	1 : M	FACT_CLAIMS	DIAG_SK	Type 1 — no SCD
DIM_FACILITY	FACILITY_SK	1 : M	FACT_CAPACITY	FACILITY_SK	Bed/occupancy daily snapshot
DIM_DRUG	DRUG_SK	1 : M	FACT_DRUG_SPEND	DRUG_SK	Always join IS_CURRENT=TRUE
DIM_PHYSICIAN	PHYSICIAN_SK	1 : M	FACT_DRUG_SPEND	PHYSICIAN_SK	Always join IS_CURRENT=TRUE
DIM_DATE	DATE_SK	1 : M	FACT_CLAIMS	DATE_SK	Enable time intelligence
DIM_DATE	DATE_SK	1 : M	FACT_CAPACITY	DATE_SK	Daily grain join

SCD Type 2 — How to Join Correctly

SCD2 dimensions have multiple rows for the same natural key (one per version). Always add the `IS_CURRENT = TRUE` filter to get only the active version. To get historical data at a specific date, use the date-range filter instead.

```
-- ★ CORRECT: join to current version of a SCD2 dimension
SELECT fc.claim_id, dp.insurance_tier, df.facility_name
FROM   SILVER.FACT_CLAIMS fc
JOIN   SILVER.DIM_PATIENT dp ON fc.PATIENT_SK = dp.PATIENT_SK
      AND dp.IS_CURRENT = TRUE -- ★ always add this
JOIN   SILVER.DIM_FACILITY df ON fc.FACILITY_SK = df.FACILITY_SK
      AND df.IS_CURRENT = TRUE; -- ★ always add this

-- ★ HISTORICAL: find the patient's insurance tier at claim date
JOIN   SILVER.DIM_PATIENT dp ON fc.PATIENT_SK = dp.PATIENT_SK
      AND fc.CLAIM_DATE >= dp.EFF_FROM -- ★ date-range
join
      AND fc.CLAIM_DATE <= dp.EFF_TO;   -- ★ captures
correct version

-- ★ VIEW HISTORY: all versions of one patient
SELECT PATIENT_ID, INSURANCE_TIER, EFF_FROM, EFF_TO, IS_CURRENT
FROM   SILVER.DIM_PATIENT
WHERE  PATIENT_ID = 'faac724a-a9e9-be66-fe1e-3044dc0ba8ea'
ORDER BY EFF_FROM; -- shows all versions oldest → newest
```

Gold-to-Materialized View Relationships

Gold Table	Key Columns	Cardinality	Materialized View	Aggregated By	Power BI Use
GOLD_CLAIMS_SUMMARY	CLAIM_DATE, FACILITY_NAME	M : 1	MV_CLAIMS_DAILY_DENIAL	Date + Facility	Denial rate KPI
GOLD_PATIENT_OUTCOMES	YEAR, MONTH, DRG_CODE	M : 1	MV_READMISSION_TREND	Week + Facility	Readmission trend
GOLD_DRUG_ANALYTICS	BRAND_NAME, GENERIC_NAME	M : 1	MV_DRUG_COST_RANK	Drug (all time)	Drug leaderboard

Quick Field Reference — Cross-File Join Map

Use this table when writing joins. Each row is one matching pair of fields across two files or tables.

Left table / file	Left column		Right table / file	Right column	Join purpose
File 1 — Drug Spend	Prscrbr_NPI	=	Silver.DIM_PHYSICIAN	npi	Prescriber details
File 2 — FHIR Patient	patient.id	=	Silver.DIM_PATIENT	patient_id	Patient demographics
File 2 — FHIR Claim	diagnosis.code	=	Reference.ICD10_CODES	ICD10_CODE	Diagnosis name lookup
File 3 — Hospital Cost	Provider CCN	=	Silver.DIM_FACILITY	provider_ccn	Facility master + SCD2
File 4 — ICD-10 TXT	ICD10_CODE	=	Silver.DIM_DIAGNOSIS	icd10_code	Diagnosis enrichment

— End of Data Dictionary & Relationship Reference —

All column definitions derived from actual downloaded files — not documentation